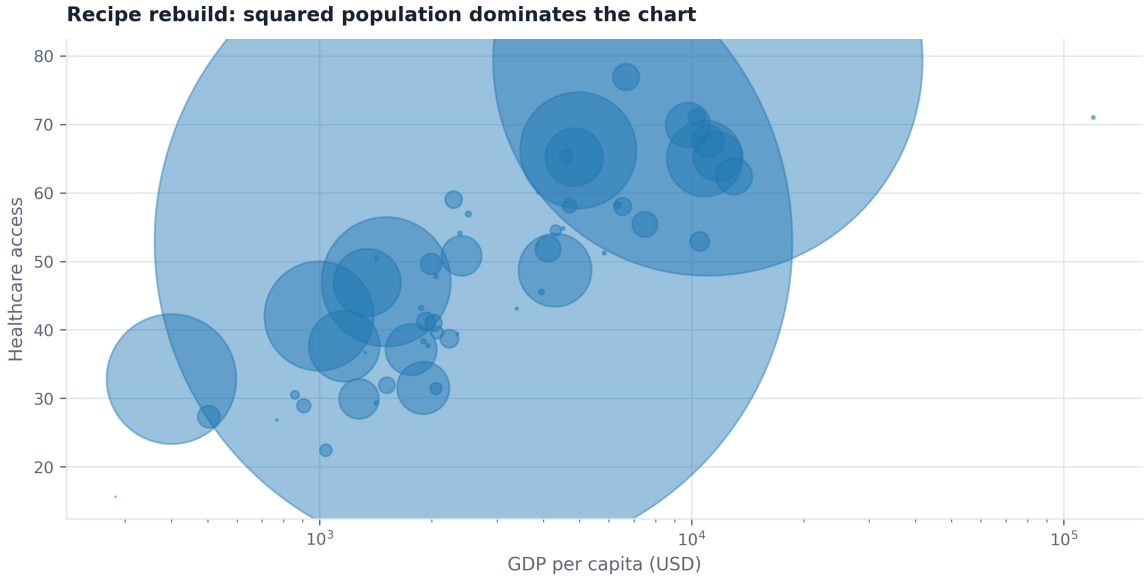
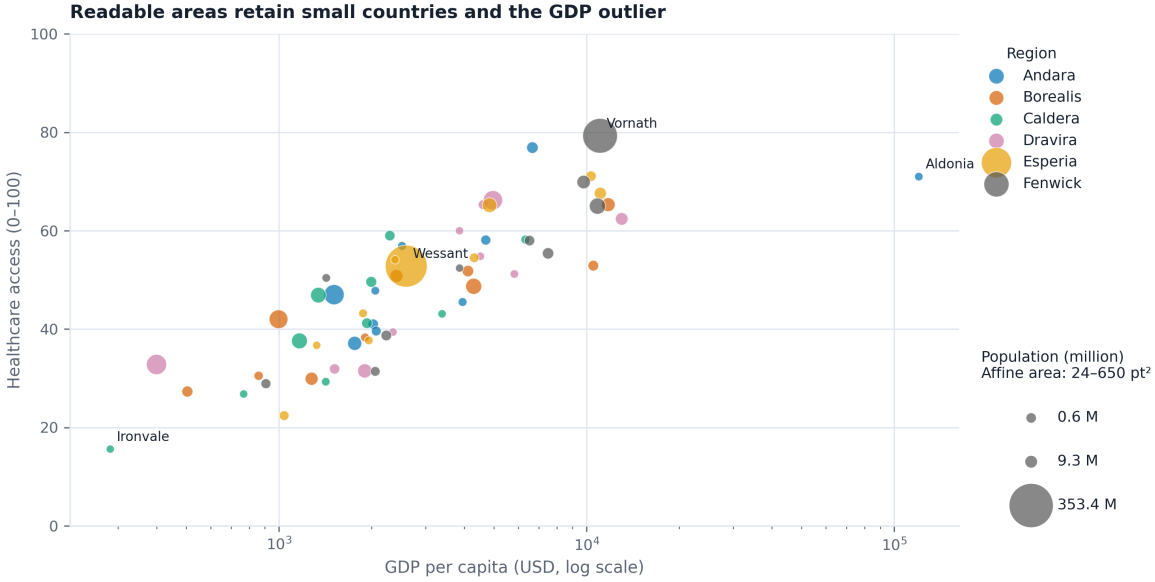
Part A | Case 1

Conclusion: population seems unrelated to access because large marks appear everywhere. Decoy: a linear GDP axis is not the cause; the recipe already specifies log GDP. Mechanism: s is area, so squaring population produces an area ratio of 346,921, not the brief's 589. The latter is the population ratio. Only 3 marks are below one point-squared; 9 are below 2 M. The supplied self-check mixes these thresholds. Redesign: bounded affine areas with a calibrated legend, log GDP and a full table for exact values; the bounded range is a readability trade-off, not proportional population area.



Synthetic CDB601220 Week 5 data; supplied boundaries, year unspecified; accessed 28 Oct 2026.
Deliberately misleading recipe: s = population² (million²); no usable size legend in recipe.

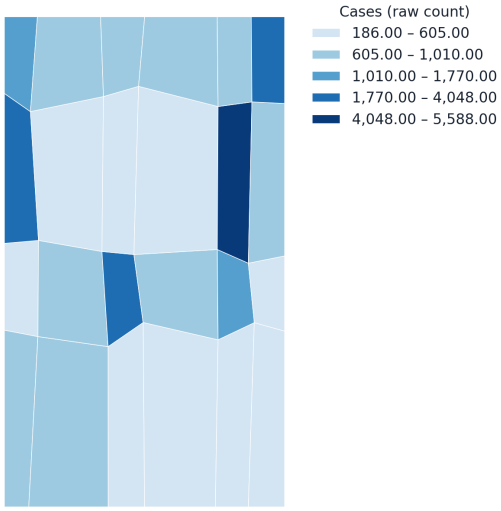


Synthetic CDB601220 Week 5 data; supplied boundaries, year unspecified; accessed 28 Oct 2026.
Outlier retained on log x. Affine area 24–650 pt² compresses population ratios; no removals.

Part A | Case 2

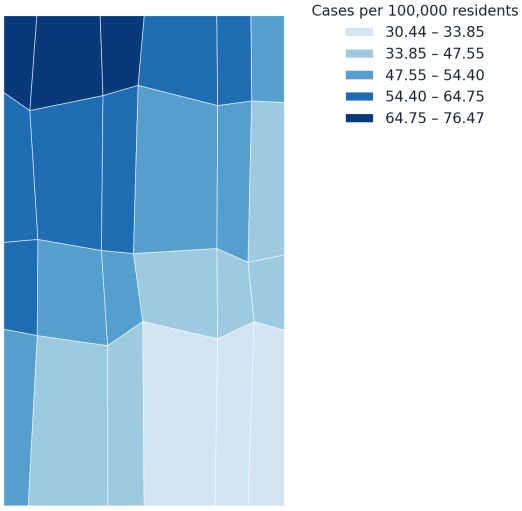
Conclusion: the capital-area districts have the greatest outbreak risk. Decoy: failed joins cannot explain this; all 24 codes match and all counts are retained. Mechanism: population is the hidden denominator. Counts correlate with population at r=0.986. Count leaders D17, D13 and D09 are replaced by rate leaders D19, D20 and D21. Counts remain relevant for workload; rates answer relative risk. Total: 30,564 cases in 57,857,000 residents.

Recipe: populous districts dominate counts



Synthetic CDB601220 Week 5 data; supplied boundaries, year unspecified; accessed 28 Oct 2026.
Lambert cylindrical equal-area; shared reference latitude. No units suppressed unless stated.

Rates identify a different set of districts

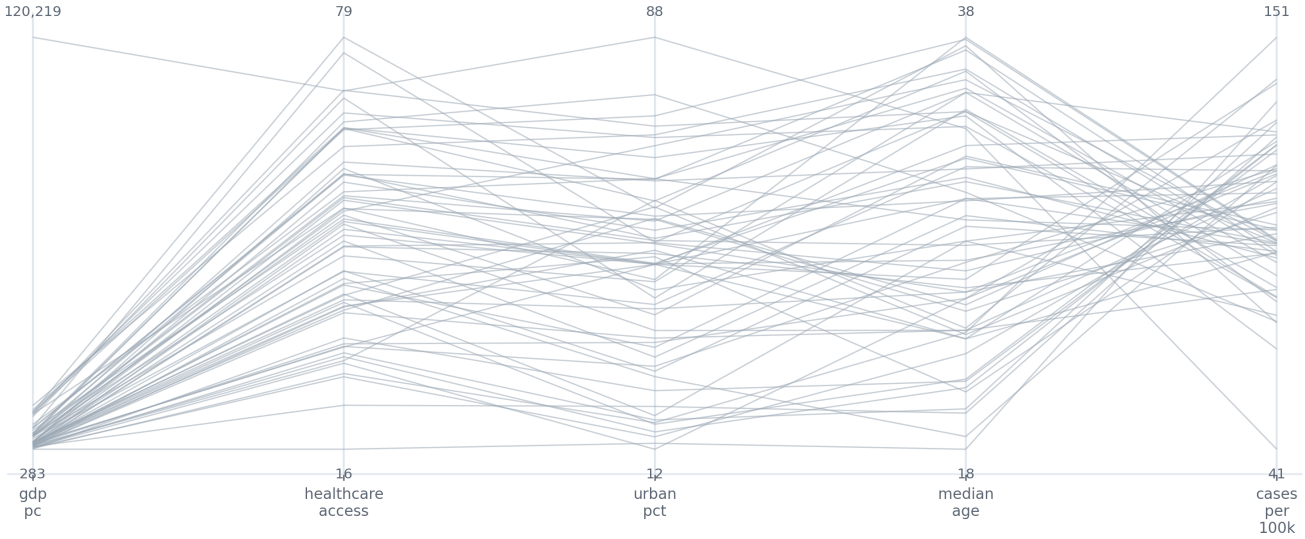


Synthetic CDB601220 Week 5 data; supplied boundaries, year unspecified; accessed 28 Oct 2026.
Lambert cylindrical equal-area; shared reference latitude. No units suppressed unless stated.

Part A | Case 3

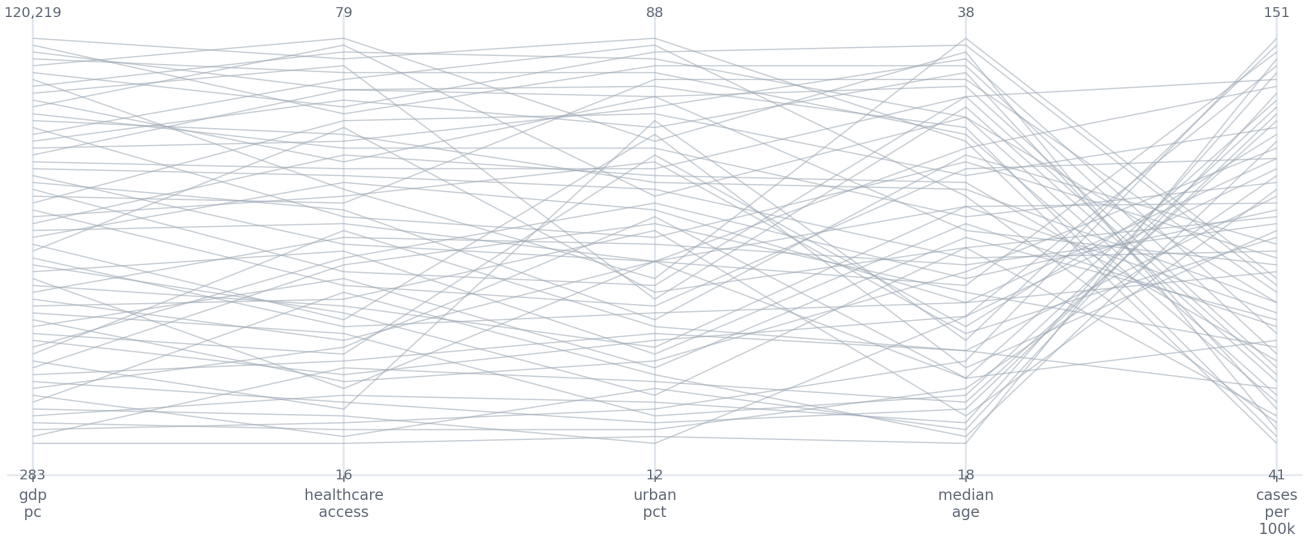
Conclusion: GDP hardly varies. Decoy: GDP does not have a narrow raw range; it spans $283–$120,219. Mechanism: the largest observation controls the min–max denominator, leaving 58 of 60 countries below 0.10 and the median at 0.017. Rank scaling makes order visible without deleting the outlier, while sacrificing distances between observations.

Recipe: min–max GDP looks almost flat; file axis order



Synthetic CDB601220 Week 5 data; supplied boundaries, year unspecified; accessed 28 Oct 2026.
All five axes min–max scaled; all 60 countries retained.

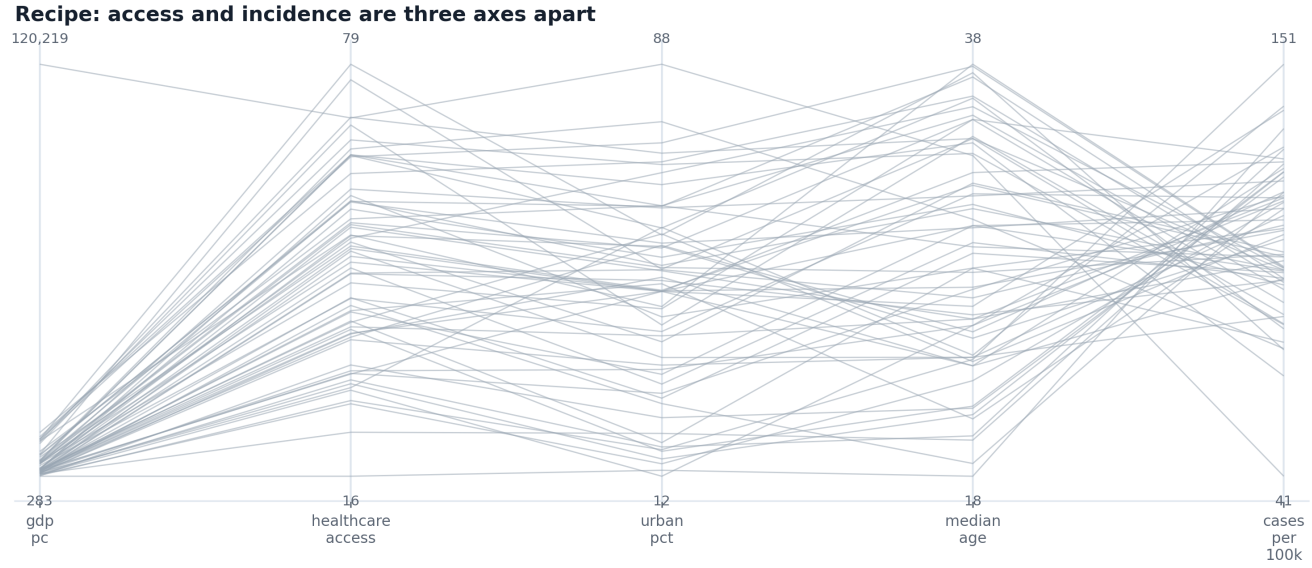
Percentile ranks reveal GDP ordering; file axis order



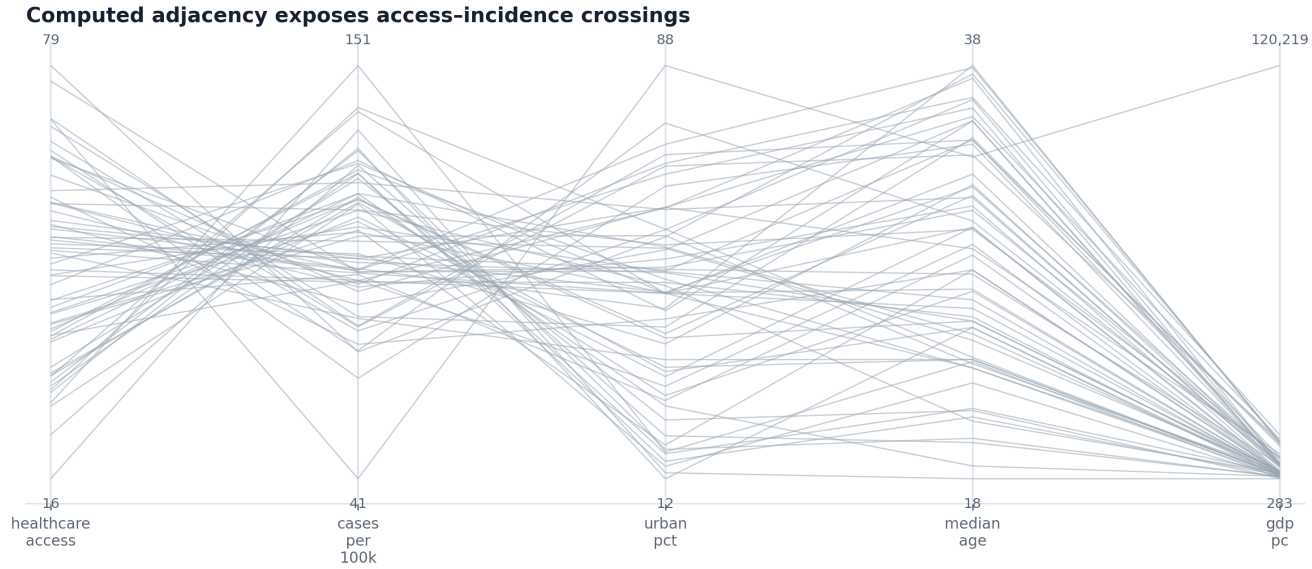
Synthetic CDB601220 Week 5 data; supplied boundaries, year unspecified; accessed 28 Oct 2026.
Rank scaling discards gap magnitude; no brushing, to isolate the scaling change.

Part A | Case 4

Conclusion: access and incidence are unrelated. Decoy: lack of numerical association is ruled out by r=-0.799. Mechanism: their file-order positions differ by three; a crossing can only be read locally between adjacent axes. The computed greedy order puts them together while holding min–max scaling constant. The inverse relationship now produces an X-shaped waist.



Synthetic CDB601220 Week 5 data; supplied boundaries, year unspecified; accessed 28 Oct 2026.
Min–max scaling; file order: gdp_pc → healthcare_access → urban_pct → median_age → cases_per_100k

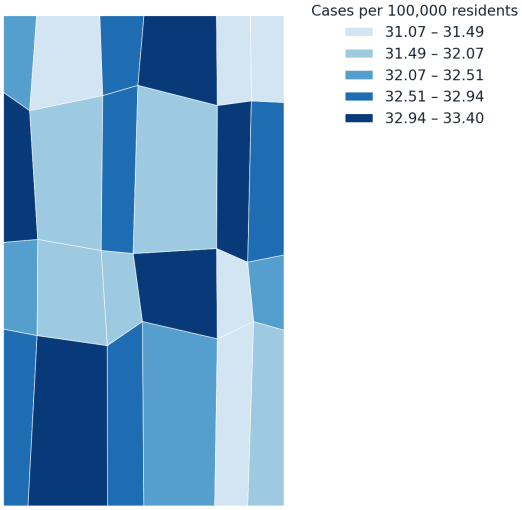


Synthetic CDB601220 Week 5 data; supplied boundaries, year unspecified; accessed 28 Oct 2026.
Min–max scaling unchanged; greedy absolute Pearson correlation order: healthcare_access → cases_per_100k → urban_pct → median_age → gdp_pc

Part A | Case 5

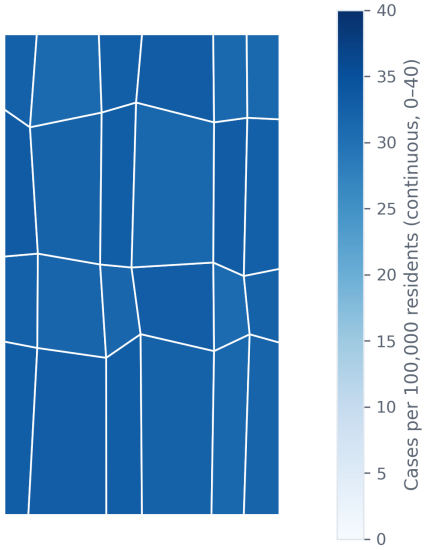
Conclusion: five distinct incidence levels form a strong gradient. Decoy: the sequential blue palette is appropriate; rainbow hue is not the problem. Mechanism: quantiles force approximately equal occupancy across five colours even though rates span 31.07–33.40, only 7.2% of the mean. A fixed continuous 0–40 scale preserves the small magnitude. Class membership is not evidence of five real risk regimes.

Recipe: five quantile bins amplify a narrow spread



Synthetic CDB601220 Week 5 data; supplied boundaries, year unspecified; accessed 28 Oct 2026.
Lambert cylindrical equal-area; shared reference latitude. No units suppressed unless stated. Five quantile classes.

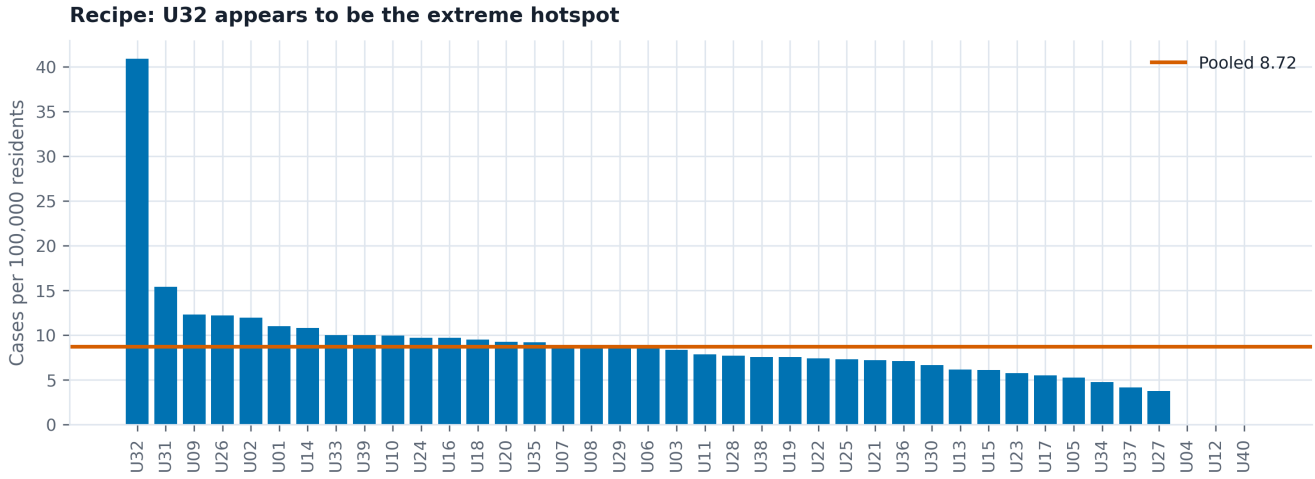
Incidence varies only slightly around 32 per 100,000



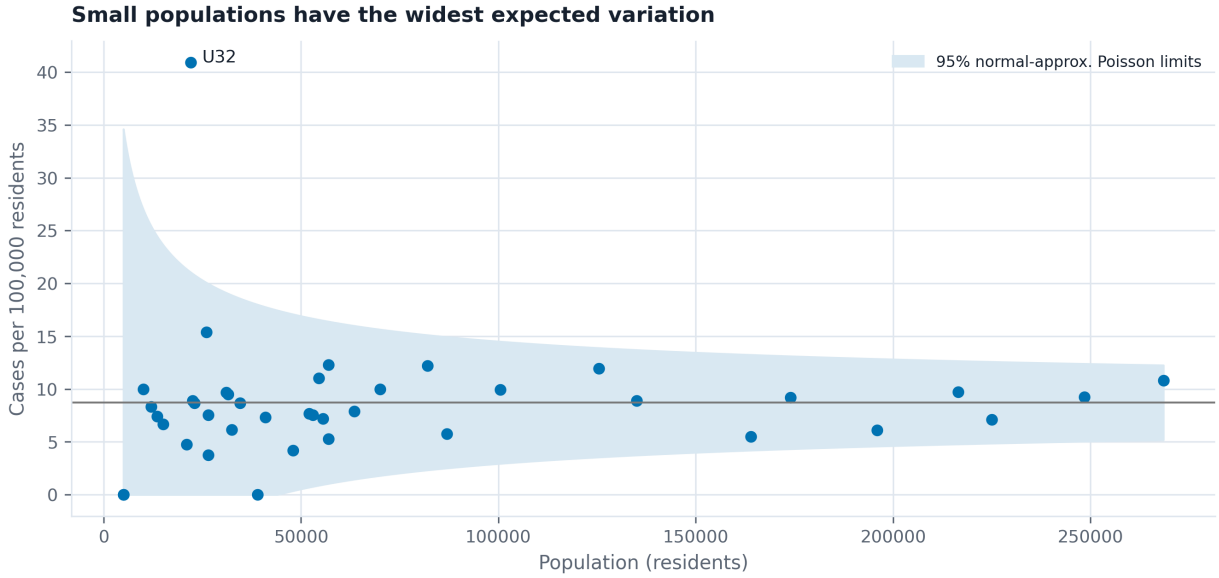
Synthetic CDB601220 Week 5 data; supplied boundaries, year unspecified; accessed 28 Oct 2026.
Lambert cylindrical equal-area; shared reference latitude. No units suppressed unless stated. Fixed zero-based 0–40 rate scale; no classification.

Part A | Case 6

Conclusion: U32 has almost five times the true risk. Decoy: the arithmetic is correct: 9/22000 × 100,000 = 40.9, versus pooled 8.72. Mechanism: random-count variance divided by a small population gives highly unstable rates. The data dictionary explicitly states every unit has the same true rate. U32 has only nine observed cases. A funnel communicates expected uncertainty; even an outlier warrants investigation rather than a causal hotspot label. No observations are removed from this diagnostic chart.



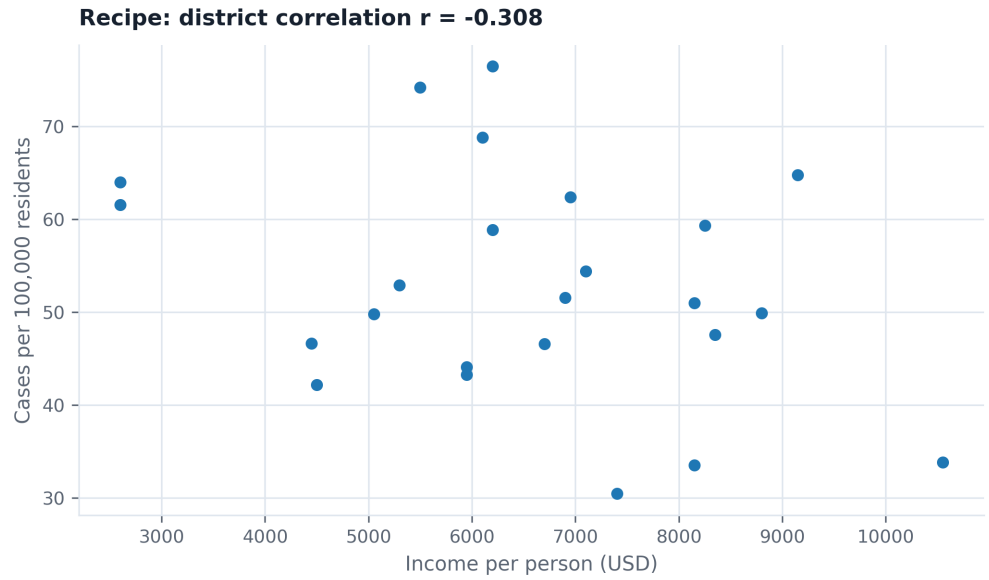
Synthetic CDB601220 Week 5 data; supplied boundaries, year unspecified; accessed 28 Oct 2026.
All 40 units; unsuppressed diagnostic exercise.



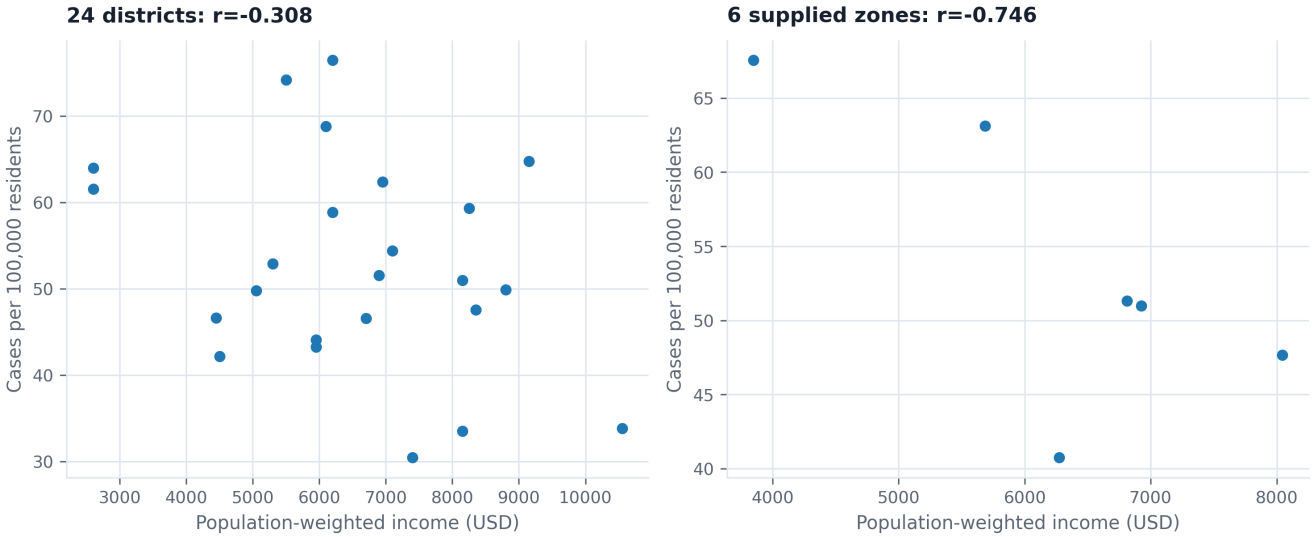
Synthetic CDB601220 Week 5 data; supplied boundaries, year unspecified; accessed 28 Oct 2026.
All units retained for stability audit; do not interpret rank as true risk. Approximate limits, not proof of equality.

Part A | Case 7

Conclusion: income and incidence are intrinsically weakly related. Decoy: mismatched totals are ruled out; aggregation conserves all 30,564 cases and 57,857,000 residents. Mechanism: the modifiable areal unit problem (MAUP). Correlation changes from -0.308 for 24 districts to -0.746 for six supplied zones. Rates are recomputed from pooled counts, never averaged unweighted. The alternate 24-unit GeoJSON has no corresponding observations or crosswalk; exact alternative-zone rates cannot be inferred without an allocation assumption. No areal interpolation is presented as measured data, and neither aggregate correlation establishes individual risk.



Synthetic CDB601220 Week 5 data; supplied boundaries, year unspecified; accessed 28 Oct 2026.
24 original districts, no removals.

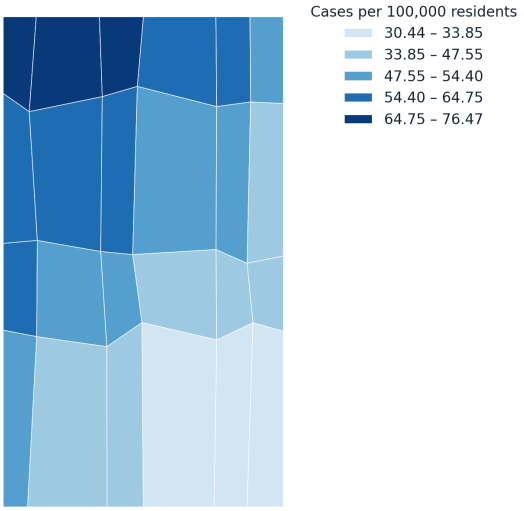


Synthetic CDB601220 Week 5 data; supplied boundaries, year unspecified; accessed 28 Oct 2026.
MAUP scale sensitivity; cases/population summed, income population-weighted. Same total population and cases.

Part A | Case 8

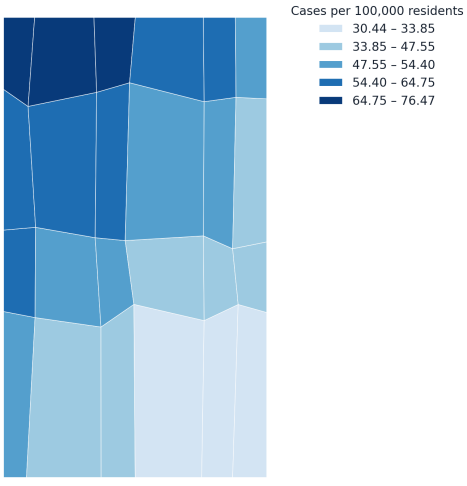
Conclusion: a rural outbreak dominates because much of the map is dark. Decoy: raw counts are not the cause here; this recipe already maps rates correctly. Mechanism: large rural polygons occupy visual area disproportionate to their population. The five largest populations account for 12.6% of land and 59.9% of residents. A rate map plus a zero-based sorted count bar separates risk from workload. Darkness over land does not establish the fraction of patients living there.

Recipe: dark land looks like most of the outbreak

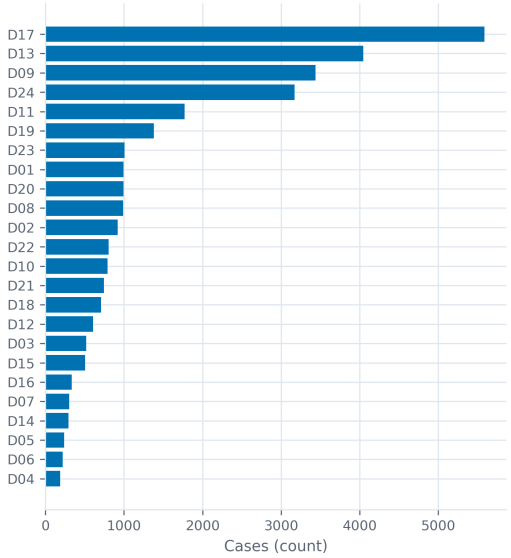


Synthetic CDB601220 Week 5 data; supplied boundaries, year unspecified; accessed 28 Oct 2026.
Lambert cylindrical equal-area; shared reference latitude. No units suppressed unless stated.

Rate shows local incidence



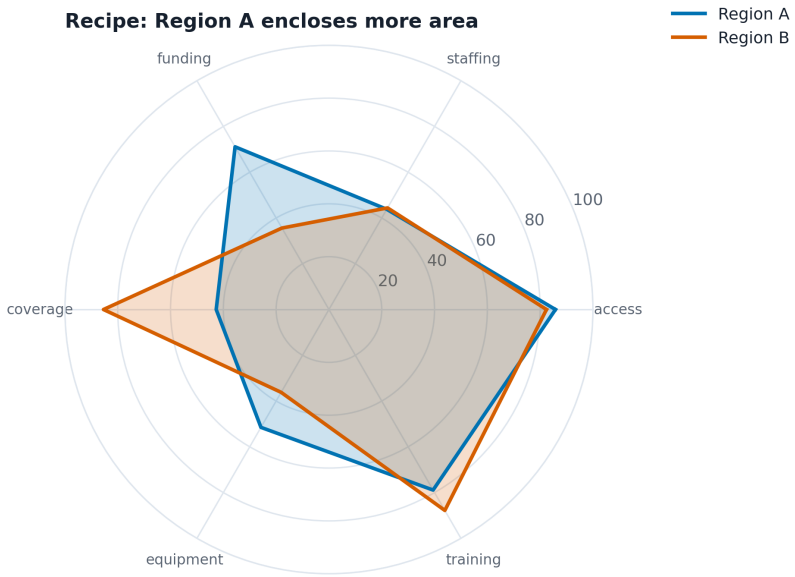
Count shows service workload



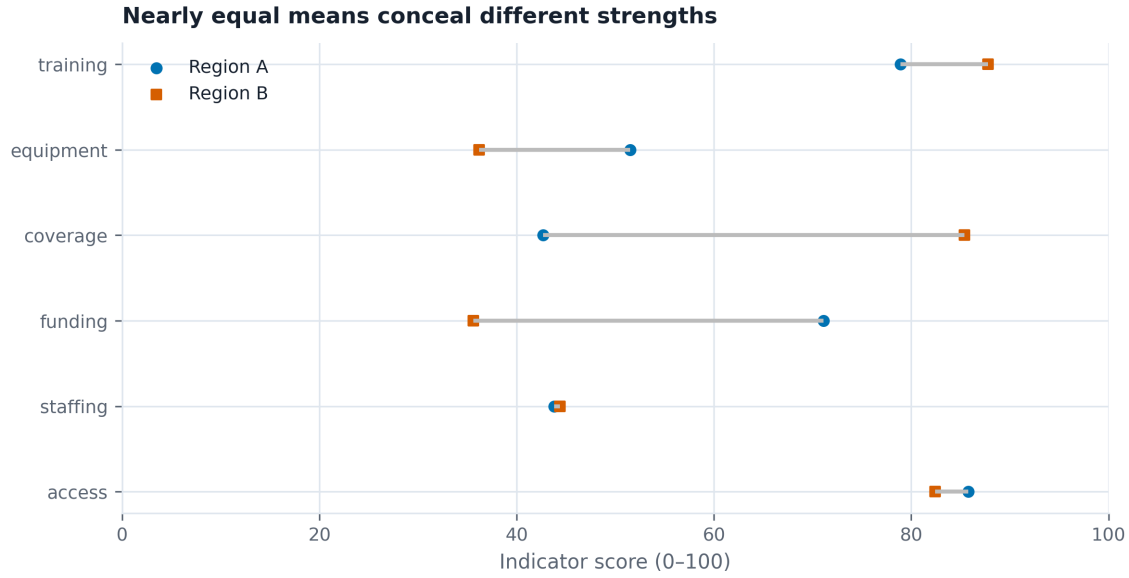
Synthetic CDB601220 Week 5 data; supplied boundaries, year unspecified; accessed 28 Oct 2026.
Lambert cylindrical equal-area; shared reference latitude. No units suppressed unless stated.

Part A | Case 9

Conclusion: Region A is stronger overall. Decoy: different measurement scales are ruled out; all six scores share 0–100, with means 62.30 and 61.97. Mechanism: radar area sums adjacent products, so arbitrary order changes area without changing any score. Original areas are 9,933 and 9,433; over 120 rotation-fixed orders A-minus-B ranges from -1,131 to 1,659, reversing the apparent winner. Aligned dots permit accurate per-indicator comparison; a single overall winner requires explicit weights.



Synthetic CDB601220 Week 5 data; supplied boundaries, year unspecified; accessed 28 Oct 2026.
Six indicators in original file order; 0–100 on each axis.



Synthetic CDB601220 Week 5 data; supplied boundaries, year unspecified; accessed 28 Oct 2026.
File indicator order; shared position scale replaces polygon area.